

Day(33):

* String Handling part 2

- we know that, on str data we can perform both indexing and slicing operations.
- By using Indexing Concept, we can get single value from str object.
- By using slicing operations, we can get range of value from str object.
- Along with indexing and slicing operations, we can perform various operations on str data by using pre-defined functions present in str object.

* pre-defined functions in str object :-

① Capitalize()

syntax: str obj.capitalize()
(OR)

str obj = str obj.capitalize()

⇒ This function is used for capitalizing the first letter first word of a given sentence only.

examples

```
>>> s = "python"
```

```
>>> print(s, type(s)) — — — python <class 'str'>
```

```
>>> s.capitalize() — — — 'Python'
```

```

>>> s = "python is an oop lang"
>>> print(s, type(s)) - - - python is an oop lang. <class 'str'>
>>> s.capitalize() - - - python is an oop lang.

```

② title()

syntax: s.title()
 (OR)
 s = s.title()

⇒ This is used for obtaining title case of given sentence
 (OR) making all words first letter are capital.

example:

```

>>> s = "python is an oop lang"
>>> print(s, type(s)) - - - same
>>> s.capitalize() - - - 'python is an oop lang.'
>>> s.title() - - - Python Is An Oop Lang.

```

③ index()

syntax: strobj.index(value)
 (OR)
 index value = strobj.index(value),

⇒ this function obtains index of the specified value.

⇒ if the specified value does not exist then we get value error.

examples

```

>>> s = "python"
>>> s.index("p") - - - 0
>>> s.index("y") - - - 1
>>> s.index("N") - - - value error.
>>> s.index("k") - - - value error.

```

④ upper()

syntax: strobj.upper()
 (OR)
 strobj = strobj.upper(),

⇒ it is used for converting any type of str data into upper case.

Examples:-

```

>>> s = "python"
>>> print(s) - - - - - same.
>>> s.upper() - - - - - 'PYTHON'
>>> s = "python is an oop lang"
>>> print(s) - - - - - same
>>> s.upper() - - - - - 'PYTHON IS AN OOP LANG'
>>> s = "123"
>>> s.upper() - - - - - '123'

```

⑤ lower()

syntax: `strobj.lower()`
 (or)
`strobj = strobj.lower()`

→ it is used converting any type of str data into lowercase.

Example:-

```

>>> s = "Data science"
>>> print(s) - - - Data science.
>>> s.lower() - - - data science.

```

⑥ isupper()

syntax: `strobj.isupper()`

⇒ This function returns True provided the given str object data is purely uppercase otherwise it returns false.

Example

```

>>> s = "PYTHON"
>>> s.isupper() - - - T
>>> s = "python"
>>> s.isupper() - - - F.
>>> s = "pYthon"
>>> s.isupper() - - - F

```

"123" or special characters is also (F).

⑦ islower()

syntax: strobj.islower()

=> This function returns True provided the given str object data is purely lower case otherwise it returns false.

examples:-

```
>>> s = "python"
>>> s.islower() — — — T
>>> s = "PYTHON"
>>> s.islower() — — — F.
```

⑧ isalpha() Alpha ^{bet}~~between~~

syntax: strobj.isalpha()

This function return True provided str object contains purely Alphabets otherwise returns False

examples:-

```
>>> s = "Ambition"
>>> s.isalpha() — — — T
>>> s = "Ambition 123"
>>> s.isalpha() — — — F.
>>> s = "AaBzDeF" — — — F.
>>> s.isalpha() — — — T.
```

⑨ isdigit()

syntax: strobj.isdigit()

=> This function returns True provided given str object contains purely digits otherwise returns false.

examples:-

```
>>> s = "python"
>>> s.isdigit() — — — F
>>> s = "123"
>>> s.isdigit() — — — T.
```


⑩ isalnum()

syntax: strobj.isalnum()

=> This function returns True provided strobj contains either Alphabets OR Numeric or Alpha Numeric only otherwise it returns False.

Example:

```

>>> s = "python310"
>>> s.isalnum() --- T
>>> s = "python"
>>> s.isalnum() --- T
>>> s = "310"
>>> s.isalnum() --- T
>>> s = ' $python310'
>>> s.isalnum() --- F

```

⑪ isspace()

syntax: strobj.isspace()

=> This function returns True provided strobj contains purely space otherwise it returns false.

Example:

```

>>> s = " "
>>> s.isspace() --- T
>>> s = ""
>>> s.isspace() --- F
>>> s = "python prog"
>>> s.isspace() --- F

```

⑫ split()

syntax: strobj.split("delimiter")
 (or)
 strobj.split()
 (or)
 listobj = strobj.split("delimiter")
 (or)
 listobj = strobj.split()

=> This function is used for splitting the given str object data into different words base specified delimiter (- _ # % ^ ^ , ; etc.)

=> The default delimiter is space.

=> The function returns splitting data in the form of list object.

examples:

```
>>> s = "Python is an oop lang".
>>> print(s)      - - - - - same.
>>> s.split()     - - - - - ['Python', 'is', 'an', 'oop', 'lang']
>>> len(s.split()) - - - - - 5
>>> x = s.split()
>>> print(x, type(x))  same <class 'list'>
>>> len(x) - - - - - 5
```

```
>>> s = "12-09-2022"
>>> print(s)      - - - - - same
>>> s.split("-")  - - - - - ['12', '09', '2022']
>>> s = "12-09-2022"
>>> dob = s.split("-")
>>> print(dob, type(dob)) - - - - - same <class 'list'>
>>> print("Day", dob[0]) - - - - - Day 12
```



```
>>> print("Month", dob[1]) - - - Month 09.
>>> print("Year", dob[2]) - - - Year 2022.
```

```
=
>>> s = "Apple # Banana # Kiwi # Guava"
>>> words = s.split("#")
>>> print(words) - - - ['Apple', 'Banana', 'Kiwi', 'Guava']
>>> words = s.split("/")
>>> print(words) - - - ['Apple', 'Banana', 'Kiwi', 'Guava']
```

(13) join()

syntax : strobj.join(iterable object)

~~Example~~ => This function is used for combining or joining list of values from any iterable object.

Examples:-

```
>>> lt = ["HYD", "BANG", "AP", "DELHI"]
>>> print(lt, type(lt)) - - - ['HYD', 'BANG', 'AP', 'DELHI'] <class 'list'>

>>> s = ""
>>> s.join(lt) - - - 'HYDBANGAPDELHI'

>>> s = " "
>>> s.join(lt) - - - 'HYD BANG AP DELHI'

>>> t = ("Rossum", "is", "father", "of", "Python")
>>> print(t, type(t)) - - - ('Rossum', 'is', 'father', 'of', 'Python') <class 'tuple'>
                                     same
>>> k = ""
>>> k.join(t) - - - 'RossumisfatherofPython'.
```

(14) starts with () :-

=> The starts with () function returns True if the string starts with specified value, otherwise false

examples

```
>>> s = "python is an oop lang"
>>> s.startswith("python") --- T
>>> s.startswith("pythm") --- F.
```

(15) ends with () :-

=> The ends with () Function returns True if the string ends with the specified value, otherwise false.

Examples

```
>>> s = "python is an oop lang"
>>> s.endswith("python") --- F
>>> s.endswith("lang") --- T.
```

(16) swapCase()

=> Make the lower case letters upper case and the upper case letters lower case.

example

```
>>> s = "PyThon"
>>> s.swapCase() --- pYtHON
=.
```

String Handling (part 2)

- predefined functions in str object are:-
- 1.capitalize()
- 2.title()
- 3.index()
- 4.upper()
- 5.lower()

- 6.isupper()
- 7.islower()
- 8.isalpha()
- 9.isdigit()
- 10.isalnum()
- 11.isspace()
- 12.split()
- 13.join()
- 14.startawith()
- 15.endswith()
- 16.swapcase()

1.capitalize():-

- syntax:- strobj.capitalize()
- or
- strobj=strobj.capitalize()

```
In [1]: s="python"
        print(s,type(s))
        s.capitalize()
```

python <class 'str'>

Out[1]: 'Python'

```
In [2]: s="python is an oop lang"
        print(s,type(s))
        s.capitalize()
```

python is an oop lang <class 'str'>

Out[2]: 'Python is an oop lang'

```
In [3]: s="pyhton" "is" "an" "oop" "lang"
        print(s,type(s))
        s.capitalize()
```

pyhtonisanooplang <class 'str'>

Out[3]: 'Pyhtonisanooplang'

2.title()

- syntax:- s.title()
- or
- s=s.title()

```
In [4]: s="python is an oop lang"
        print(s,type(s))
        s.capitalize()
        s.title()
```

python is an oop lang <class 'str'>

Out[4]: 'Python Is An Oop Lang'

3.index()

- syntax:- strobj.index(value)
- or
- indexvalue=strobj.index(value)

```
In [6]: s="python"
        s.index("p")
```

Out[6]: 0

```
In [7]: s.index("y")
```

Out[7]: 1

```
In [8]: s.index("n")
```

Out[8]: 5

```
In [9]: s.index("N")
```

ValueError

Traceback (most recent call last)

Cell In[9], line 1

-----> 1 s.index("N")

ValueError: substring not found

4.upper()

- syntax:- strobj.upper()
- or
- strobj=strobj.upper()

In [10]:

s="python"
print(s)

python

In [11]:

s.upper()

Out[11]: 'PYTHON'

In [12]:

s="pyhton is a oop language"
print(s)
s.upper()

pyhton is a oop language

Out[12]: 'PYHTON IS A OOP LANGUAGE '

In [13]:

s="123"
print(s)
s.upper()

123

Out[13]: '123'

5.lower()

- syntax:strobj.lower()
- or
- strobj=strobj.lower()

In [15]:

s="DATA Science"
print(s)
s.lower()

DATA Science

Out[15]: 'data science'

6.isupper():

- syntax:- strobj.isupper()

In [16]:

s="PYTHON"
s.isupper()

Out[16]: True

In [17]:

s="python"
s.isupper()

Out[17]: False

In [18]:

s="PYThon"
s.isupper()

Out[18]: False

In [19]:

s="123"
s.isupper()

Out[19]: False

7.islower()

- syntax:- strobj.islower()

In [20]:

s="python"
s.islower()

Out[20]: True

```
In [21]: s="PYthon"
s.islower()
```

Out[21]: False

8.isalpha()

- syntax:- strobj.isalpha()

```
In [22]: s="Ambition"
s.isalpha()
```

Out[22]: True

```
In [23]: s="Ambition 123"
s.isalpha()
```

Out[23]: False

```
In [24]: s="AsBrtyERTY"
s.isalpha()
```

Out[24]: True

9.isdigit()

- syntax:- strobj.isdigit()

```
In [25]: s="python"
s.isdigit()
```

Out[25]: False

```
In [26]: s="123"
s.isdigit()
```

Out[26]: True

10.isalnum()

- syntax:- strobj.isalnum()

```
In [27]: s="python310"
s.isalnum()
```

Out[27]: True

```
In [28]: s="python"
s.isalnum()
```

Out[28]: True

```
In [29]: s="310"
s.isalnum()
```

Out[29]: True

```
In [30]: s="$ python310"
s.isalnum()
```

Out[30]: False

11.isspace()

- syntax:- strobj.isspace()

```
In [31]: s=""
s.isspace()
```

Out[31]: False

```
In [32]: s=" "
s.isspace()
```

Out[32]: True

```
In [33]: s="python prog"
s.isspace()
```

Out[33]: False

12.split()

- syntax:- strobj.split("delimiter")
- or
- strobj.split()
- or
- listobj=strobj.split("delimiter")
- or
- listobj=strobj.split()

```
In [34]: s="pyhton is an oop language"
print(s)
s.split()
```

pyhton is an oop language

Out[34]: ['pyhton', 'is', 'an', 'oop', 'language']

```
In [35]: len(s.split())
```

Out[35]: 5

```
In [40]: x=s.split()
print(x,type(x))
len(x)
```

['pyhton', 'is', 'an', 'oop', 'language'] <class 'list'>

Out[40]: 5

```
In [46]: s="12-09-2022"
print(s)
s.split("-")
```

12-09-2022

Out[46]: ['12', '09', '2022']

```
In [49]: s="12-09-2022"
dob=s.split("-")
print(dob,type(dob))
```

['12', '09', '2022'] <class 'list'>

```
In [50]: print("day",dob[0])
```

day 12

```
In [51]: print("month:",dob[1])
```

month: 09

```
In [52]: print("year:",dob[2])
```

year: 2022

```
In [55]: s="apple#banana#kiwi/guava"
words=s.split("#")
print(words)
```

['apple', 'banana', 'kiwi/guava']

```
In [56]: words=s.split("/")
print(words)
```

['apple#banana#kiwi', 'guava']

13.join()

- syntax:- strobj.join(iterable_object)

```
In [58]: lst=["hyd","bang","ap","delhi"]
print(lst,type(lst))
```

['hyd', 'bang', 'ap', 'delhi'] <class 'list'>

```
In [61]: s=""
s.join(lst)
```

Out[61]: 'hydbangapdelhi'

```
In [62]: s=" "
s.join(lst)
```

Out[62]: 'hyd bang ap delhi'


```
In [64]: t=("rosum","is","father","of","python")
         print(t,type(t))

('rosum', 'is', 'father', 'of', 'python') <class 'tuple'>

In [65]: k=" "
         k.join(t)

Out[65]: 'rosum is father of python'
```

14.startswith()

- syntax:- strobj.startswith(name)

```
In [68]: s="python is an oop language"
         s.startswith("python")

Out[68]: True

In [70]: s.startswith("Python")

Out[70]: False
```

15.endswith()

- syntax:-strobj.endswith(name)

```
In [71]: s="python is an oop lang"
         s.endswith("python")

Out[71]: False

In [72]: s.endswith("lang")

Out[72]: True
```

16.swapcase()

- syntax:-strobj.swapcase()

```
In [73]: s="PYThon"
         s.swapcase()

Out[73]: 'pytHON'

In [ ]:
```